LESSON PLAN

AIR WEAPON (10)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	12	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Film show demonstrating fluid flow characteristics
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Film show demonstrating fluid flow characteristics.
5. Specific Objectives	10.1.1.1 Trainee will be able to explain the properties of atmosphere, flow over bodies, boundary layer and various concepts of Aerodynamics.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Nil
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary properties of the atmosphere affecting flight?
(b) Define the boundary layer in simple terms.
(c) List the four main forces acting on an aircraft.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Properties of Atmosphere • Flow over Bodies • Boundary Layer Concepts • Aerodynamic Principles	L & Ds	PPT, IB	15 min
(b)	• Film Analysis: Flow Patterns • Visualizing Boundary Layers • Understanding Pressure Distributions • Key Aerodynamic Concepts	L & Ds	PPT, PDF	20 min
(c)	• Summary of Fluid Characteristics • Real-world Applications • Review of Aerodynamic Terms • Connection to Future Lessons	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the boundary layer affects drag on an aircraft body.
(b) Describe the relationship between fluid flow characteristics and lift generation.

12. Home Assignment and Follow Up	Prepare a brief note on how the concepts of fluid flow and boundary layers discussed in this lesson relate to the geometrical characteristics of aerofoils covered in LP 09.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Geometrical characteristics of aerofoil, definition of terms NACA series, various forces e.g. lift, drag and moments, theories of Lift production center of pressure, aero dynamic center, force and pitching moment, coefficient of lift, variation of co-efficient of lift with aspect ratio and sweep back.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min